Sections: B,C & D

Linear Algebra (MT-1004)

Homework#32

Instructor: Muhammad Ali

Hat Matrix

Question 1

The Hat matrix, denoted as H, is defined as:

$$H = X(X^T X)^{-1} X^T$$

where X is the design matrix in a linear regression model. Answer the following questions about its properties:

(a) **Symmetry:** Prove that the Hat matrix H is symmetric, i.e., $H = H^T$.

(b) **Idempotence:** Show that the Hat matrix is idempotent, meaning $H^2 = H$.

(c) **Trace:** Demonstrate that the trace of H equals the number of predictors (including the intercept term), i.e., $\text{Trace}(H) = p$.

(d) **Leverages:**

  (a) Define the diagonal elements of H, h_{ii}, known as leverages.

  (b) Show that $0 \leq h_{ii} \leq 1$ for all i, and that $\sum_{i=1}^{n} h_{ii} = p$.

(e) **Residuals:**

  (a) Prove that the residuals $e = y - \hat{y}$ satisfy $He = 0$, where $\hat{y} = Hy$.

  (b) Show that $\hat{y}$ and the residuals e are orthogonal.

(f) **Eigenvalues:** Discuss the eigenvalues of the Hat matrix. Why are they restricted to the values 0 and 1?